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import nltk

import pandas as pd

from nltk.tokenize import word_tokenize

from nltk.corpus import stopwords

from nltk.stem import PorterStemmer, WordNetLemmatizer

from sklearn.feature_extraction.text import TfidfVectorizer

# Download necessary NLTK data

nltk.download('punkt')

nltk.download('punkt_tab')

nltk.download('stopwords')

nltk.download('averaged_perceptron_tagger_eng')

nltk.download('wordnet')

# 1. Sample Document

document = "The curious cats are catching small mice in the dark garden."

# 2. Tokenization (Breaking sentences into words)

tokens = word_tokenize(document.lower())

# 3. Stop Words Removal (Removing common words like 'the', 'is', 'in')

stop_words = set(stopwords.words('english'))

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filtered_tokens = [w for w in tokens if w not in stop_words and w.isalnum()]

# 4. POS Tagging (Identifying Parts of Speech: Nouns, Verbs, etc.)

pos_tags = nltk.pos_tag(tokens)

# 5. Stemming (Chopping off ends of words)

stemmer = PorterStemmer()

stems = [stemmer.stem(w) for w in filtered_tokens]

# 6. Lemmatization (Reducing word to its dictionary root)

lemmatizer = WordNetLemmatizer()

lemmas = [lemmatizer.lemmatize(w, pos='v') for w in filtered_tokens]

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7. TF-IDF Representation

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corpus = [document, "Mice are small animals.", "The garden is dark."]
vectorizer = TfidfVectorizer()
tfidf_matrix = vectorizer.fit_transform(corpus)

# Display TF-IDF as a Table
df_tfidf = pd.DataFrame(tfidf_matrix.toarray(), columns=vectorizer.get_feature_names_out())

print("Original Tokens:", tokens)
print("POS Tags:", pos_tags[:5])
print("Lemmatized:", lemmas)
print("\nTF-IDF Matrix:\n", df_tfidf)
```